Mathematical Formulation of Smart Physiotherapy Rehab System using XGBoost

1 Raw Sensor Input (Time Series)

From the IMU device (MPU6050), the raw signal at time t is:

$$X_t = \big[a_x(t), a_y(t), a_z(t), g_x(t), g_y(t), g_z(t)\big]$$

where:

- a_x, a_y, a_z represent linear acceleration
- g_x, g_y, g_z represent angular velocity

The full dataset is a time series:

$$\{X_1, X_2, X_3, \ldots, X_n\}$$

2 Sensor Fusion and Angle Computation

To convert raw IMU data into joint angle, a complementary filter is used:

$$\theta_t = \alpha\big(\theta_{t-1} + g_t \cdot \Delta t\big) + (1-\alpha)\theta_t^{acc}$$

where:

$$\theta_t^{acc} = \tan^{-1}\left(\frac{a_y}{a_z}\right)$$

- g_t is gyroscope reading
- $\alpha \in [0, 1]$ is the fusion coefficient (typically ≈ 0.98)

Thus, we obtain the angle signal:

$$\theta(t)$$

3 Feature Extraction

From the angle signal $\theta(t)$, the following features are computed:

3.1 Range of Motion

$$ROM = \max(\theta) - \min(\theta)$$

3.2 Angular Velocity

$$\omega(t) = \frac{d\theta}{dt}$$

$$\omega_{avg} = \frac{1}{n}\sum_{t=1}^{n} \omega(t)$$

$$\omega_{max} = \max(\omega(t))$$

3.3 Stability (Variance)

$$\sigma^2 = \frac{1}{n}\sum_{t=1}^{n} \left(\theta_t - \bar{\theta}\right)^2$$

3.4 Jerk (Smoothness)

$$jerk = \frac{d^2\theta}{dt^2}$$

3.5 Time per Repetition

$$T = t_{end} - t_{start}$$

3.6 Final Feature Vector

$$\mathbf{x} = \left[\max(\theta),\, \min(\theta),\, ROM,\, \omega_{avg},\, \omega_{max},\, \sigma^2,\, jerk,\, T\right]$$

4 XGBoost Model

4.1 Prediction Function

The final prediction is computed as:

$$\hat{y} = \sum_{k=1}^{K} f_k(\mathbf{x})$$

where:

- f_k is the k-th decision tree
- K is the total number of trees

4.2 Tree Function

Each tree maps input to a leaf weight:

$$f_k(\mathbf{x}) = w_{q(\mathbf{x})}$$

where:

- $q(\mathbf{x})$ is the leaf index
- w is the leaf weight

4.3 Multi-class Classification (Softmax)

$$P(y = c \mid \mathbf{x}) = \frac{e^{\hat{y}_c}}{\sum_j e^{\hat{y}_j}}$$

5 Objective Function

The model minimizes:

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$$

where:

- l is the loss function (e.g., log loss)
- $\Omega(f)$ is the regularization term

5.1 Regularization

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum w^2$$

where:

- T is number of leaves
- w are leaf weights

6 End-to-End Mathematical Flow

$$X_t \to \theta(t)$$

$$\theta(t) \to \mathbf{x}$$

$$\mathbf{x} \to \hat{y} = \sum f_k(\mathbf{x})$$

$$\hat{y} \to y \in \{0, 1, 2\}$$

7 Key Insight

Time Series → Feature Engineering → Tabular ML

XGBoost does not operate directly on raw sensor signals; instead, it relies on structured numerical features derived from the signal.

8 References

References

[1] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2016. [Online]. Available: `https://arxiv.org/abs/1603.02754`

[2] Wearable Sensor-Based Rehabilitation Exercise Assessment. [Online]. Available: `https://www.researchgate.net/publication/273953220`